Response to scientific review

Gregory Morse Peter Rakyta

7 November 2026

We thank the reviewer for independently checking the identities, finite-alphabet search, invariant inequality, and lower-bound argument. The revision follows the theory-paper direction. Its title is now *CNOT costs of controlled balanced involutions and an invariant two-SQiSW criterion*. The atlas is an accompanying reference resource. All seven numbered concerns have prompted changes to the text, data presentation, experiments, or claim scope.

1. Grid size, obstruction types, and optimality denominators

We agree that the headline grid size hid the distribution of difficulty. The manuscript now reports the named targets and fixtures separately and gives one complete table per phase policy, broken down by physical width. The 217 named one-qubit entries, 192 phased Cliffords, and standard determinant and discrete-alphabet obstructions are explicitly identified. Zero-entangler constructions are separated from positive-entangler constructions.

A new row-resolved CSV reconciles every retained cell with its explicit matched proof sources. We take a union rather than adding overlapping audits; fixed-target minima and minimum uniform family budgets are disjoint columns. Legacy Boolean optimality flags alone are not counted as certificates. Among 2,295 positive-entangler literal constructions, the union contains 732 fixed-target minima and 710 uniform family minima; 853 remain unproved under this accounting. Among 3,057 effective positive-entangler projective constructions, it contains 984 fixed-target minima and 955 uniform family minima; 1,118 remain unproved. These are entangler certificates, not local-cost optimality claims.

The reported 1,916 projective denominator counts explicitly stored projective results. A further 1,141 positive-entangler literal circuits were reused projectively, giving the complete-grid denominator 3,057. The old raw flags 694/423 and selected-audit counts were different sets and were inadequately explained; neither is presented as the new union. The new CSV records both the raw flag and the matched proof sources so this distinction can be checked.

The obstruction table identifies all 2,601 literal determinant-group cases. All 268 projective obstructions occur in the three discrete presets. These are described as standard feasibility tests, with exceptional parameter specializations distinguished from whole-family representation.

2. Clifford reference and missing atlases

We removed any framing of the 192 one-qubit targets as a major unexplored synthesis class. Matsumoto–Amano, Kliuchnikov–Maslov–Mosca, and Giles–Selinger are now cited. The finite-alphabet shortest-path result is presented as an explicit reference and regression certificate, with no continuous-alphabet global-local-optimality extrapolation. We did not add a two-qubit Clifford classification or imply that it has been completed.

The complete gate and Clifford PDFs and their editable sources are now included in the full review ZIP, rather than only pointed to elsewhere. The manuscript's coverage tables carry the main quantitative evidence; atlas page count is not used as evidence of algorithmic novelty.

3. Prior art, generalization, and proof detail

We added Yu–Duan–Ying, Yu–Ying, Shende–Markov, and Nielsen et al., and explain how their lower-bound and controlled-product arguments relate to ours. Ordinary Fredkin, Toffoli, and controlled fermionic SWAP have different relative blocks and resource counts. Unrestricted two-qubit gates are kept separate from Ising-class interactions.

The main new generalization classifies every controlled balanced two-qubit involution: $V = V^\dagger$, $V^2 = I$, $\operatorname{tr} V = 0$. A Pauli-coefficient normal form proves costs four at OSR four, three at OSR two, and one or two for local products according to support. It includes the zero-vector and rank-deficient degeneracies and gives exact symbolic boundary constructions. There is no rank-three stratum. We separately prove that nonlocal balanced relative blocks need three unrestricted two-qubit gates and provide a three-gate upper construction. The fermionic SWAP example therefore costs four Ising interactions but only three unrestricted interactions.

We do not assert the suggested classification of arbitrary controlled-V by V's Weyl class alone: I, Z_a, and Z_aZ_b have the same local Weyl class but controlled costs zero, one, and two. The corrected generalization uses the additional involution and support structure explicitly.

The four-Ising proof now defines control OSR, derives the target cut ranks, counts all twelve surviving interaction words, and expands the conjugation bound into $M_0 \otimes I + M_1 \otimes Z$. Fermionic SWAP's Pauli expansion shows OSR four directly and its local equivalence to iSWAP is stated. The improved literal witness has four CNOTs and six Ry rotations, replacing the former ten-local-gate witness. Exact tests cover the free two-angle normal form and every rank-two boundary, not only numerical samples.

4. Baselines and application comparison

We executed Qiskit 2.6.2, pytket 2.18.5, and BQSKit 1.2.1 on eight independent fixed targets, using both library-source and matrix routes where supported. There are 42 SDK attempts. We additionally executed Quartz and QUESO on the eight source circuits each, preserving raw outputs, revisions, equivalence rules, harness changes, budgets, and fixed-wire full-matrix replay. Their objective is total NAM gates; only one-qubit fusion changes their reported local alphabet. UniversalQCompiler remains unrun because its licensed kernel is unavailable; that explanation is not applied to the open-source tools.

BQSKit matches four CNOTs and six locals for fFredkin. Our separate 108-layout BQSKit experiment finds a four-CNOT fit in about six seconds. These numerical findings support the reviewer's observation that the upper bound is easy; the lower proof and generalized classification carry the mathematical contribution. All eleven older SDK entangler wins are now explicitly identified as fFredkin across different presets. Compiler losses are also reported, including Quartz's verified 30-CNOT C4X result versus the current 36-CNOT atlas entry, and smaller C3X/SSWAP local counts from other tools. No claim of universal dominance remains.

The tomography-figure 27-to-3 comparison is removed from the active manuscript and evidence appendix. The latter's incomplete reference is replaced with Warren et al.'s full title, authors, venue, year, and DOI. Published C3X/C4X and ordinary Fredkin constructions are discussed with

matching ancilla and resource assumptions and are identified as upper bounds, not automatically global minima.

The parameter-reuse table now prominently includes independent numerical recompilation: all ten bound instances give 18 CNOTs, 40 locals, depth 25, at a mean of about 7.68 ms. This beats the symbolic 18/42/31 circuit in local count and depth. The older 36-CX gatewise lowering is explicitly a weak contextual control, and the known three-CX fSim source is acknowledged. We withdraw the inference of a new 36-to-18 resource advance over analytical fSim synthesis. Cold synthesis, SDK compile time, and parameter binding are separated.

5. OSR ablation and practical utility

We added a construction ablation disabling OSR factorization and residual peeling while retaining native charts, Clifford/Pauli construction, and posthoc verification. It gives the same two-qubit success count as the full method, 11/12, with one deadline each. The prior chart-disabled result is 3/12. The added opaque three- and four-qubit product probes remain unresolved for both methods in this configuration. These outcomes are retained and reported; they do not demonstrate an OSR-search advantage on this workload. The title no longer attributes these chart successes to OSR-guided search. OSR's role in the theorems is established by proof instead.

We also implemented a bounded atlas peephole pass on eight existing benchmark sources in two native presets. Every proposed replacement requires exact identity acceptance; the control is the same gatewise native lowering. The pass finds no improving windows on the suite. A separately labeled three-CX SWAP positive control checks its exact identity and native count accounting. We report this negative suite outcome and make no general peephole speedup claim. This is evidence about the current optimizer's limits, not hidden or counted as successful application improvement.

6. Secondary score and symbolic-expression cost

The parameter-description score and its dependence on simplification are moved to an appendix. It is not a calibrated pulse model and does not enter the main entangler theorems. The historical negative-cycle repair is no longer part of the scientific contribution. Unknown equal-entangler scores remain incomparable.

The long SQiSW chart is now called a machine-inspectable scalar DAG, not a short directly readable expression. A new optional compiled numerical evaluator shares dependencies and preserves lazy degeneracy branches. For the 585-binding chart, compilation takes about 1.36 s and 1,000 warm bindings about 0.97 s in the recorded run. Full matrix replay passes at 64 points, including singular corners and angle folds. This checks practical reuse of an already exactly certified family, without claiming that it always beats numerical KAK or that sampling proves an identity. The text explains when symbolic identity and phase contracts are useful and explicitly retains numerical specialization as a competitive alternative.

7. Presentation and editorial cleanup

The manuscript is theory-first, with contributions stated directly and scope limits consolidated. It names Gregory Morse and Peter Rakyta, removes anonymous-review labeling, cites every displayed figure in the text, and expands the control-rank argument. Affiliations and individual human contribution statements remain for the authors to supply before submission; none has been invented.

We removed coordination prose from the proof supplement and scientific archive view. Original historical records remain immutable in the repository; the current review package uses the cleaned scientific view and current sources. The new response, evidence appendix, source bundles, complete atlases, and verification manifests describe the revised claims consistently. AI assistance is disclosed as assistance, not authorship or proof authority.

Additional SQiSW result

We proved the iff statement suggested by the review. For chamber coordinates, write $g_1 = (u+iv)^2$ with $u \geq 0$; then $|g_1| + \Re g_1 = 2u^2$. The two invariant inequalities are equivalent to $W \geq 4u$. Their difference factors as

$$W - 4u = 2[\cos 2x - \cos 2(y + |z|)][\cos 2x - \cos 2(y - |z|)].$$

The chamber orders fix the sign of the second factor; $z = 0$ is handled separately. This proves equivalence to $x \geq y + |z|$, including the closed boundary, and Huang et al.'s known region gives necessity and sufficiency. We credit the region to Huang et al. and claim the invariant reformulation and checked implementation, not a new pulse-cap theorem.